

Docker Task - Sudir

Task Description:

Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)

Techstacks needs to be used :

- AWS EC2
- Docker

How do I submit my work?

- Push all your work files to GitHub (O/P screenshot images must).
- Submit your URLs in the portal.

Terms and Conditions?

- You agree to not share this confidential document with anyone.
- You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.
- We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

1. Update the System

sudo apt update -y

2. Install Docker

sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y

Screenshot:

```
sudo apt update -y
sudo apt install docker-ce docker-ce-cli containerd.io -y
```

3. Verify Docker Installation

docker --version

docker info

Screenshot:

```
root@sonarjen:~# docker --version
Docker version 28.5.0, build 887030f
root@sonarjen:~# docker info
Client: Docker Engine - Community
Version: 28.5.0
Context: default
Debug Mode: false
Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.29.1
    Path: /usr/libexec/docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
    Version: v2.40.0
    Path: /usr/libexec/docker/cli-plugins/docker-compose
Server:
  Containers: 0
    Running: 0
    Paused: 0
    Stopped: 0
  Images: 0
  Server Version: 28.5.0
  Storage Driver: overlay2
    Backing Filesystem: extfs
    Supports d_type: true
    Using metacopy: false
    Native Overlay Diff: true
    userxattr: false
  Logging Driver: json-file
  Cgroup Driver: systemd
  Cgroup Version: 2
  Plugins:
    Volume: local
    Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
  CDI spec directories:
    /etc/cdi
    /var/run/cdi
  Swarm: inactive
  Runtimes: io.containerd.runc.v2 runc
  Default Runtime: runc
  Init Binary: docker-init
  containerd version: b98a3aace656320842a23f4a392a33f46af97866
```

4. Pull a Docker Image

`docker pull nginx`

Screenshot:

```
root@sonarjen:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
8c7716127147: Pull complete
250b90fb2b9a: Pull complete
5d8ea9f4c626: Pull complete
58d144c4badd: Pull complete
b459da543435: Pull complete
8da8ed3552af: Pull complete
54e822d8ee0c: Pull complete
Digest: sha256:f79cde317d4d172a392978344034eed6dff5728a8e6d7a42f507504c23ecf8b8
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
```

5. List Docker Images

`docker images`

Screenshot:

```
root@sonarjen:~# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    07ccdb783875   14 hours ago   160MB
```

6. Run a Docker Container

`docker run -d -p 8081:80 --name web nginx`

- `-d` → Detached mode
- `-p 8081:80` → Map host port 8081 to container port 80
- `--name web` → Container name

Screenshot:

```
root@sonarjen:~# docker run -d -p 8081:80 --name web nginx
ce86841fcf2a2621ddcb704ebc1fc9ddba58616744398f8e9d83b5a3bdlb5d62
```

7. List Running Containers

`docker ps`

`docker ps -a`

Screenshot:

```
root@sonarjen:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
ce86841fcf2a   nginx    "/docker-entrypoint..." 9 seconds ago  Up 9 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp  web
root@sonarjen:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
ce86841fcf2a   nginx    "/docker-entrypoint..." 16 seconds ago Up 15 seconds  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp  web
```

8. Stop and Start Container

`docker stop web`

`docker start web`

Screenshot:

```
root@sonarjen:~# docker stop web
web
root@sonarjen:~# docker start web
web
```

9. Create and Inspect a Docker Volume

`docker volume create mydata`

`docker volume ls`

`docker inspect mydata`

- Volume used to persist data

Screenshot:

```
root@sonarjen:~# docker volume create mydata
mydata
root@sonarjen:~# docker volume ls
DRIVER      VOLUME NAME
local      mydata
```

10. Run Container with Volume

`docker run -d -v mydata:/usr/share/nginx/html nginx`

- Mount volume mydata to /usr/share/nginx/html in container

Screenshot:

```
root@sonarjen:~# docker volume create mydata
mydata
root@sonarjen:~# docker volume ls
DRIVER      VOLUME NAME
local       mydata
root@sonarjen:~# docker run -d -v mydata:/usr/share/nginx/html nginx
928c885ad7ab61db270275ef9532629eaae8b32e7dade7575622d8ad86ac7cc8
root@sonarjen:~# docker inspect mydata
[
  {
    "CreatedAt": "2025-10-08T10:55:49Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/mydata/_data",
    "Name": "mydata",
    "Options": null,
    "Scope": "local"
  }
]
```

11. List Docker Networks

`docker network ls`

Screenshot:

```
root@sonarjen:~# docker network ls
NETWORK ID      NAME      DRIVER      SCOPE
1889c9c492e5    bridge    bridge      local
5f0f60426c11    host      host        local
efdlf51ab7c7    none      null        local
```

12. Create a Custom Network

`docker network create mynetwork`

`docker run -d --name container1 --network mynetwork nginx`

`docker inspect mynetwork`

- Create mynetwork and attach container
- Inspect network to view container IP, subnet, etc.

Screenshot:

```
root@sonarjen:~# docker network ls
NETWORK ID      NAME      DRIVER  SCOPE
1889c9c492e5    bridge    bridge  local
5f0f60426c11    host      host    local
efdlf51ab7c7    none      null    local
root@sonarjen:~# docker network create mynetwork
f511afdd6634a9e755d5ca61d09ad37a70cf357e04609f7854dded895d3ba006
root@sonarjen:~# docker run -d --name container1 --network mynetwork nginx
4dd0cc10326138798d2ac1a152d31ee572534f8fbd6278a929169829459328e2
```

```
root@sonarjen:~# docker inspect mynetwork
[
  {
    "Name": "mynetwork",
    "Id": "f511afdd6634a9e755d5ca61d09ad37a70cf357e04609f7854dded895d3ba006",
    "Created": "2025-10-08T10:56:28.633721189Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.18.0.0/16",
          "Gateway": "172.18.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "4dd0cc10326138798d2ac1a152d31ee572534f8fbd6278a929169829459328e2": {
        "Name": "container1",
        "EndpointID": "00e57050e3a6cf977c919a71920e1d1f3f9017b9f7f2db1ef276c1a6fa5c3bf8",
        "MacAddress": "0a:55:84:b3:11:ba",
        "IPv4Address": "172.18.0.2/16",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]
```

13. Remove Docker Images or Containers (Optional)

docker stop web

docker rm web

docker rmi nginx

Screenshot:

```
root@sonarjen:~# docker stop web
web
root@sonarjen:~# docker rm web
web
root@sonarjen:~# docker rmi nginx
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2e:6e:4d brd ff:ff:ff:ff:ff:ff
    altname enx0800272e6e4d
    inet 192.168.0.99/22 metric 100 brd 192.168.3.255 scope global dynamic enp0s3

root@sonarjen:~# docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:f79cde317d4d172a392978344034eed6dff5728a8e6d7a42f507504c23ecf8b8
Deleted: sha256:07ccdb7838758e758a4d52a9761636c385125a327355c0c94a6acff9babff938
Deleted: sha256:71b75b17511f67932ccf71e2046c6dlb4fe17a594134c765bf71c874dedc7027
Deleted: sha256:c76da83ebfb3e35184dla7105d03cecfb144d07cb4dae12378392040fbd44615
Deleted: sha256:f825da5ac2d31a2c717db4fa159b6728b33a94bcb7285bdldfebcbbe23ebdl85
Deleted: sha256:31333e0f1ecf5940213f8b1ed4eb3e4d78d77fa5fb59clcd05a6672690edl33c
Deleted: sha256:e72eb931613f8clb5baf39f864877beee0780af9f9d20c5a02790d0146f0b012
Deleted: sha256:775e3183eed457c31a0855ac7a55b6e0cb8d40554e9524cb6d503cb2351e0b10
Deleted: sha256:ld46119d249f7719e1820e24a311aa7c453f166f714969cffe89504678eaa447
root@sonarjen:~#
```

Output:

Note:

This task is done in my local VM using oracle VB and Ubuntu os
my bridge adapter IP

```
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2e:6e:4d brd ff:ff:ff:ff:ff:ff
    altname enx0800272e6e4d
    inet 192.168.0.99/22 metric 100 brd 192.168.3.255 scope global dynamic enp0s3
```

